

FANTASTIC UNIVERSE

SCIENCE FICTION

JULY
35c



WIDENITZ
KING SIZE
PUBLICATION

THREE FEATURED NOVELETS:

THE REWRITE MAN

by Ward Moore

FOUNDING FATHERS

by Robert Bloch

THINKING MACHINE

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*** START OF THE PROJECT GUTENBERG EBOOK THE THINKING
MACHINE ***

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The Thinking Machine

by **Raymond F. Jones**

Like Robert Heinlein and Harry Bates, Raymond F. Jones can point with pride to a major Hollywood production based upon one of his stories, and more science fantasy anthology inclusions over the past seven or eight years than there are blackbirds in an electronic wizard's pie. And now he takes a look at the future as excitingly prophetic as the chill alien face which stared remorselessly out at us from the silver screen in THIS ISLAND EARTH and left us wondering far into the night how long the human race would survive.

**“Leave everything to me!” the Machine seemed to whisper.
“Love and marriage, your daily bread.” But there will
always be rebels.**

They gave Rick Theron a party before he left.

It wasn't a rational thing to do, or even completely legal. But they were Agros, Sixth Rating, and it was the kind of thing to be expected among their class. The Watch Police knew it was going on, and didn't even bother to validate the report to the regional observatory.

So they darkened the windows in Sol Hanara's house and invited all the people from the neighboring farms—the friends that Rick Theron had known since he was a child—and gave a farewell party. It was a special doings to show the special kind of feeling they had for Rick.

Not that they wouldn't have liked to do the same thing for everyone who went. They would. But it was for Rick that all those accumulated and repressed desires were unloosed.

Rick's father, Sard Theron, drove one of the farm's big cargo planes down from Rillo to RiGrand. He got clearance for the flight as a cargo of livestock, but even the traffic officials knew that he was picking up a dozen other families from farms along the way. It didn't really matter. The Lonestar Region was among the top producing areas in the country. No one could expect its Agros to be civilized as well.

The big plane rolled to a stop, the last of three bringing guests to the party, and the farm families climbed down to the solid surface of the RiGrand field. They could smell the night-borne scent of the Gulf waters and see the distant shimmering of moonlight on its surface.

Rick had always loved it down here. He'd wished ever since he was a boy that his family could have transferred assignments to a farm nearer the water, but his father preferred the big, open country of Rillo for cattle and roughage crops and the gathering in of a plentiful harvest.

Now Rick was through with both kinds of country. He stood a moment at the foot of the plane stairs sucking the moist Gulf air into his lungs. This was probably the last night he'd ever see this wonderful RiGrand Agro area.

"There he is!"

He heard the sounds of voices and running feet, and saw the shapes of moving figures in the half light. They came up and swarmed over him, slapping him on the back.

“How’s it feel to have your farming days over?”

“Hear you’re picked to pair with a Mech!”

“You’ll get fed and brushed like a prize sire—what a life!”

“They say those Mech women are charged like a parogun. You have to say ‘yes’m’ and ‘no’m’ or you’re likely to get your head smoked off.”

“Lay off it!” Rick roared, half annoyed by the banter that was supposed to be friendly enough but which had the effect of small, hot needles in his skin.

A vacuum of sound swelled outward in the wake of his voice. “Sorry, guys,” he said. “I didn’t mean it to come out that way. I just don’t like it being the last night of my life at RiGrand.”

“Sure, we don’t like it, either. Guess there’s better ways of showing it than being damned fools. Let’s go in and have a tall drink.”

They crowded around, taking his arms, urging him on. These were the fellows he’d grown up with on the vast farms and ranches of the Lonestar Agro Region. They knew how he felt. Or at least they were trying. Most of them couldn’t really know. They’d never leave. They’d spend the rest of their lives here.

For some inconceivable reason the Machine had picked *him* to leave. To pair with a Mech. What kind of breed was that supposed to produce? A Rillo range sire and a salt grass heifer!

He went along into the enormous guest hall of the RiGrand ranch. The families left the drinking table and surged about him, shaking his hand and offering congratulations as if something wonderful had happened to him. It was enough to make a man think they *wanted* him to leave.

But he knew how they felt. It was just a part of growing up, to them. Maybe the trouble with him was that he didn’t want to grow up.

Mendon Carter, RiGrand’s Chief Agro, stood up on the drinking table to propose a toast. His mane of white hair was entangled with points of quivering, silver light. “To Rick Theron!” he cried. “May a thousand Agro sons call him father!”

It wasn't a good toast. Not for a man who was going to pair with a Mech. But Mendon Carter was trying, in the best way he knew how, to supply an expression of their love which Rick would have to remember and sustain him in the years ahead. Until the time, at least, when he would finally all but forget that he had ever been an Agro.

"To Rick!" the others shouted, even more loudly than they intended, in order to cover their embarrassment at Mendon's inept toast.

But Rick pretended there was no cause for discomfort. He leaped to the table beside Mendon and held his glass high. "To my sons!" he cried. "I'll send every one of them back to Lonestar!"

That eased their tension, and gave them license to be as gay as they liked. They answered back with shouts of laughter and good wishes as Rick jumped down from the table, his eyes intent upon the figure he had been searching for ever since he came in.

She was standing against the far wall, her face bright with laughter that pained him. He forced his way through the crowd to Barie's side.

"For a little while I thought you weren't here," he said.

"You knew better than that," Barie answered. "Come on." She took his arm. "The orchestra's going to play. You aren't going to dance with anyone but me tonight. And after tonight you'll never dance with me as long as you live."

She stepped closer and he took her in his arms as the squeaky, tootling orchestra assembled by Mendon Carter at one end of the room took up its beat.

II

It wasn't a good orchestra and never would be. It was just some of the farm boys who liked to get together on Saturday nights and beat out a little whimsey. Sometimes it was at RiGrand; sometimes at Rillo or Worth-Dallas. At best it was a five hundred mile round trip, and of course the Watch Police knew all about it. They didn't expect anything better from Agros, Sixth Rating.

They were doing something called "The Lass at the Mech-Shake Ball."

Barie bounced happily to its senseless, repetitious rhythm. Rick felt the warm touch and vibration of her body against him, but he couldn't keep time.

"What's the matter?" Barie said finally. "Don't you want to dance with me tonight?"

To everyone but him it seemed like a time for dancing. "I want to get out of here," he said.

He took her out to the patio under the open sky, where the stars were so sweet and clear it made something ache inside a man. They wouldn't look like that anywhere else in the world, even if you roamed the world over.

He stopped with Barie just beyond an arbor, from which they could see the distant sheet of restless water against the horizon. He touched her and drew her close to him. She looked up, smiling, and patted his cheek.

"I don't think you want to go very badly," she said.

"Am I supposed to?"

"It's like growing older every year. You may not want to, but it happens just the same."

"You mean there's nothing we can do when a machine a thousand miles or more from here says one day that Rick Theron is to leave the land he loves and go to a city he never wanted to see and there be paired with a Mech he never wanted to know?"

Barie laughed softly and pressed a little closer to him. “Rick, you talk so silly. I guess that’s why I’ve always liked you. Even the first time we met, when Dad took me with him to Rillo to talk about cattle with your father. I remember I thought I’d never seen a boy who talked like you. I remember how you spoke of the cattle and the land as if they were your very own. You were nine then. Do you remember?”

Didn’t she know? Didn’t she know he remembered every glimpse he’d ever had of her, every sound her voice had made in his hearing? He gripped her arm tightly in his big, rough hand. His eyes swept the fields and the grassy plain that led down to the sea.

“Is there anything wrong with having and owning something for your very own?” he demanded.

“Who’d want to? You’re hurting my arm, Rick.”

He released her. “I’d want to,” he said. “I’d want some land, some cattle that were mine and couldn’t be taken away. Something to love and take pride in.”

He turned her so that he could see her eyes by the light from the sky. “You don’t know what I’m talking about, even yet, do you?” he said.

Barie shook her head. “I don’t understand you, Rick. Sometimes you frighten me with your talk that I can’t make any sense out of. Why should you want to *own* something? It already belongs to you. You work the land. You turn in its production. You make your living here.”

“Except that now I have to go away and never come back again.”

“It makes no difference!” Barie cried. “Wherever you go you’ll have something to do, something to produce, something that will get you your living. If the Machine says you belong in another place, paired to a certain Mech, then that’s right. There’s no argument with it. The Machine can think better than any man ever could! You know that’s so.”

He regarded her for a long moment. Then swiftly he took her into his arms and pressed his mouth hard against hers. She clung to him, answering his yearning. But even in the midst of her embrace he felt a coldness somewhere that was like a little stab of terror in his mind.

He raised his face from hers. “Does the Machine know how to think about love?” he said softly.

“It must.” Barie shrugged and straightened in his arms. “It says a certain Mech girl is more right for your arms than I am.”

“And don’t you *hate* it for saying that?”

“Hate it for knowing more than our poor, stupid brains do? Hate it for being right? Would it make sense if you and I went ahead and made a dreadful mistake that would bring us misery all our lives?”

“Barie, do you really believe that? In your heart—?”

“I think I almost hate *you* when you pretend to be so blind,” Barie said. “You know it’s true as well as I do.”

“Barie—Barie—has it all been so empty? Has it all been meaningless—since that day when I was nine and showed you what land and cattle could really mean to a man?”

He had to turn away to keep her from sensing the sudden, quick sob that welled up from the depths of him. And then she came to him and stroked his long hair with her soft fingers.

“Of course it has not been meaningless and empty,” she said. “It’s been wonderful. We’ve been happy. Our childhood has been sweeter because of each other. And now it is time to grow up. Do you want childhood to last forever, Rick?”

He stood looking down at her face—at her long black hair teasing about her shoulders in the soft night wind and her wide dark eyes so full of trust for the world to which they were born that she would not listen even to her own heart.

It added to the smouldering, uncertain hate in his own heart against that same world.

“Is it childishness that we love each other?” he demanded almost roughly.

For a moment the expression of calm acceptance on her face seemed to break. “It must be,” she whispered at last, “because no one keeps it. We have everything else; we mustn’t expect to keep childhood. And even if we wanted to, what could we do about it?”

“We can go away! Listen to me, Barie. There hasn’t been much time to plan, but I’ve been thinking of this ever since the notification came. Tonight’s our chance.”

“What are you talking about? Go where?”

“The Machine doesn’t do the thinking for the whole world. There are savages it doesn’t care about. We could be one with them!”

“How?”

He pointed to the dark shapes of the planes beyond the farmhouse. “I’ve planned for us to take one of them. We can fill it with spare tanks of fuel and fly it as far south as the Andes. They’d never find us there. They’d never want to!”

She seemed to hesitate a long time, watching the direction of his glance. “How would you get it ready? The escort comes for you in the morning. There isn’t time—”

“*They’ll* help me load fuel.” He nodded toward the farmhouse where the music was thumping ecstatically. “Say you’ll go with me, Barie! It’s our only chance!”

“All right,” she whispered. “If you can get them to help with making the plane ready, I’ll go.”

III

Rick almost ran from her, bumping into two figures standing at a corner of the house, smoking. It was Len and Sam from his own farm.

"It's not fair," Len complained jokingly, "to spend all your time with Barie. After all, you *are* going to be paired with a Mech in a couple more days!"

"I'm not!" he cried in exultation. "I'm going with Barie. I need your help." In a tumble of words he told them what he planned. "Get a half dozen more guys from inside the house, Sam. Len, you come and help me get started. We'll pump the tanks full and figure out a way to feed them from spare barrels inside the plane. There ought to be enough barrels around the farm."

But neither of the men had moved. "Come on!" he said.

"You can't," said Sam. "You know better, Rick. It isn't right. The Machine's told you what's best. We like you too much to help with a fool thing like this. You and Barie—we like you both too much."

He saw it then, he thought. Sam had always liked Barie. Maybe he figured with Rick out of the way the Machine might give her to him.

"It takes something like this to find out who your friends are," Rick cried. "There are plenty of guys who *will* help me!"

"We're your friends," said Len quietly. But he was gone from hearing.

One by one, or in groups of two or three, they were approached by him in the hall or out in the yard. They were appalled by the stupidity of the thing he asked them to do. He was stunned by their repeated refusal.

In panic, he looked about him. There was still dancing in the hall, but it was muted now, and the jumping, frantic rhythm had reduced to a low beat that was more felt than heard. Their eyes were upon him, averting as he stared back in defiant bewilderment.

He had not asked them all, but word had gone around from mouth to ear, until all knew his frantic decision.

He held them in his stare when he had asked his last man, and knew it was futile to ask any longer. He whirled from the room and raced out to the hangars alone.

The night was half gone and he had only until daylight to work, but he had to try. He unreeled the fuel hose and fed it to the partially empty tanks of the plane. While it throbbed with the pump's pulsations, he ransacked the tool and supply sheds for drums to fill the cabin.

He gathered some, but there didn't seem to be nearly enough. What he had he wrestled up the landing stairs and through the doorway. He lashed them down and went searching for more, then came back to change the hose connection on the tanks.

The dancing had ceased inside the house. All the guests, those who were his friends and had known him all his life, came out and stood in silent knots watching his frantic movements silhouetted against the night sky as he tried to prepare the plane alone.

But it was too much of a task. Dawn came, and he hadn't found enough barrels. He hadn't filled those he had, and he hadn't devised the way to feed their contents to the main tanks from inside the plane.

He was licked, but he was still trying when they saw the plane of the Watch Police against the brightening sky, coming to take him to the re-orientation center.

He climbed down in defeat and looked slowly about the ring of sober faces of his friends who had watched and refused to help. "I could have done it," he cried, "if you'd given me a hand. I leave behind no friends at Lonestar—only enemies!"

And then it came to him for the first time, apparently, that Barie could have helped, but she hadn't. He turned to her, in the forefront of the group, and seemed to read in her eyes the thing he had always known was there but didn't dare to admit.

She had known it would be this way. She had never intended going with him.

"You can't keep childhood forever, Rick. I had to show you that," she said. "It was the last good thing I could do for you!"

Six days ago he'd had nothing in his heart but love for the land—and for Barie. He'd expected he'd have the right to pair with her. If you were an Agro, you could afford such hopes. The Machine didn't bother Agros too much. It let them pair often with the ones they had chosen themselves.

Only a few were denied it. And fewer still were taken away from the Region altogether, to pair and spend the rest of their lives elsewhere. Rick remembered those who had gone. Stan More, Daly Croden—not more than a couple of dozen since he'd been aware of the kind of world he lived in, and the things that sometimes happened to people.

Now, he was one of them, borne through the sky to an alien land that he had never seen, never wanted to see. It had happened to him.

He watched the unfamiliar landscape through the windows of the plane. A couple of Watch Police were his companions, but they were busy over a game and paid no attention to him. His thoughts remained with the world he was leaving behind.

There was something wrong with it. He had never been able to put his finger on it, but he had always been the odd one, the strange one who didn't fit. He wanted things of his own—in a world where each man could say that all the Earth was his, if he remembered his neighbor could say it too. He didn't believe that the great Machine, which they told him did the thinking for mankind, could think out his own problems as well as he could.

Somehow this wrongness in the world had destroyed Barie. That was the sign by which he knew it was the world and not he who was in error. It could make her stand and say it was right for someone else to take her place in his arms.

Barie—Barie—

It had taken from him every reason for living.

He had no idea what was ahead of him, but in his mind there burned a single decision that would cover anything that came up. He would do everything in his power to obstruct the goals and purposes to which the Machine assigned him. He would break, and destroy, and disobey until they either crushed him or let him go back. And he had never heard of any who went back.

With this decision clearly made, his mind was free to speculate on what he might find. He knew the traditional stories that circulated among the Agros regarding the Mech cities and their vast production centers. He'd heard tales of the monstrous, unthinkable mechanical brain that had finally brought order out of the chaos of human civilization.

The thinking machine, as the Agros called it, had direct control over the life of every human being on the face of the Earth. At birth, a tape of data based on brain and chromosome mapping was filed with the central library which fed the machine. Here it was instantly available for optimum meshing with similar data from all other men and women currently alive.

In as much detail as the machine considered necessary then, the subsequent lives of the people were planned. Their station, occupation, matings, permission to reproduce—all of these activities and thousands more, at the machine's option, were precisely dictated and controlled.

Rick was vaguely aware of the history of the changeover from chaos and war to order and peace. It had come about as a natural byproduct of the Second Industrial Revolution. Men had been searching for the answer for all the ages of history, and then suddenly they had it without even looking for it.

Automation. Robots took the place of clerks and mechanics in the factories. Machines could remember ten thousand times as many facts as the most efficient human and apply those facts without error. There was dislocation and unrest, but eventually the economy settled down just as it had in the First Industrial Revolution.

And then the Great Idea that was so obvious and yet so long delayed. Automation in government. Millions of accurate data on tape to determine the need for a new law or change in the old one. No Senator or Representative could hope to match knowledge with such a device. But the one impregnable argument held for many years: This was mere machine calculation. You couldn't depend for lawmaking on an operation in which no judgment was involved. Human beings had to be governed by human beings. There was no other way.

But finally this limitation, too, was overcome. Marcroft's invention of the judgment circuit made it possible to construct a machine capable of

duplicating human thought. The chain was complete.

The changeover was almost painless. Automation, conscription, emergency regulation had been with the world so long that it had become the normal way of life. Here at last was a device that could, without error, direct a man's life in the most useful channels, and do it for *all* men on the globe, without favoritism or bias. Here at last was the millennium for which all the prophets and seers had searched.

Only a few, like Rick Theron, felt, burning within them, aching remnants of something that was not satisfied or taken into account by the Machine.

IV

Rick Theron had never been away from the farms before in his life and his first sight of the shining city of Sanlou almost took his breath away and made him forget momentarily his hate and his resolution to defy the Machine. His longing for the fields and ranges, the patient cattle and green crops dimmed a trifle. He hadn't known a city could be beautiful, too.

The plane dropped from the sky almost at the base of its outermost tall building. Only then was he aware that the Watch Police had been observing him closely for a long time.

The plane rolled to a stop.

"Okay. This is it, farmboy," one of them said. "Let's go in and get the manure cleaned off your feet for the last time."

There was no clear impression of what followed immediately—a thousand blurred details of registration, identification, movement through pleasing corridors and rooms, a complete physical examination that included everything but a count of his body cells. Then, at the end of the day, a moment of rest in the chambers of the Orientation Officer to whom he had been assigned.

The man was tall, like Rick. He had lighter hair, but a countenance that was no less rugged, as if he knew what it was to work with his hands under the open sky.

"I'm Jackson," he said. "Bernard Jackson. We're going to work in pretty close association for the next few months, so we might as well take it as easy as possible. I know a few things about you, but I'd like to hear more. And then I'll tell you anything you want to know about your position here. Fair enough?"

Rick found himself liking the man, just as he had the city, in spite of his determination to hate everything which had taken him from the farms.

But caution warned him against confiding fully in Jackson. He told the story of his life, briefly, confining himself to those aspects of it which the

man must already know. He said nothing of Barie and his terrible disappointment in losing her.

Even this seemed to be known, however. “There was a girl, wasn’t there?” said Jackson. “A girl named Barie?”

Rick hesitated. “Yes—there was.”

“How do you feel about parting from her?”

He shrugged and then remembered Barie’s own words. “You can’t keep childhood forever.”

He repeated them, convinced that he was saying the right thing. The Orientation Officer nodded and smiled. “Sometimes we have a little trouble over such situations and corrective steps are necessary. Apparently none will be needed in your case.”

He looked directly at Rick.

“As to your own situation now the Machine has discovered an extraordinary combination of potential abilities in you, and has been able to match them in a very desirable counterpart with whom you will be paired.

“You must understand that we are not wholly unhuman and mechanical about all of this. You will be properly introduced and given adequate opportunity to become well acquainted. We have special areas set aside for these courtship preliminaries. Whenever you find it mutually agreeable an official sanction is given to your pairing and you are then ready for the next step of preparation for your place in our society. The getting acquainted period usually takes about two to three weeks.”

“What if it doesn’t become mutually agreeable?” Rick asked impulsively and wished immediately that he’d held his tongue.

Bernard Jackson was not disturbed by the question, however. He merely smiled. “That just doesn’t happen. The Machine doesn’t make mistakes.”

Rick managed to get a good night’s sleep in spite of the experiences of the day. A long time later he reminded himself that his ability to sleep had been a very fortunate thing, too—considering all that was to follow during the next twenty-four hours.

He was introduced to Deva Warel, the woman with whom he was scheduled to pair, at noon the following morning.

Bernard Jackson brought them together for lunch. She was seated at a table near the window of the dining room in the Orientation Building. With her was Jackson's counterpart, a female Orientation Officer, who seemed unhappy about the whole arrangement.

Jackson pointed her out ahead of time, as they came into the room. "That's her, sitting right over there," he said.

Rick looked. For a moment he seemed to be staring at a small, golden sunburst. Then it turned into the reddest head of hair he had ever seen or imagined.

"That's Deva? The one with the red hair?"

"In the flesh."

He knew at once that the Machine's traditional infallibility was sheer idiocy. This girl was as unlike Barie as it was possible for a woman to be, and on her face, as she caught sight of him and turned, was a look of total, unadulterated antagonism.

"She has the highest potential value of any Mech designer we have discovered for many years," Jackson whispered as they approached. "So you see, you are being entrusted with a very valuable property," he added, with a smile Rick didn't understand at all.

Introductions were quickly completed. The name of Deva's companion was Flora Johns. She nodded politely, but Deva was openly contemptuous. Inside himself, Rick felt like laughing. It occurred to him that the machine must give out a good many solutions that human beings found difficult to assimilate. This, certainly, was going to be one of the most indigestible on record. In a way, he and Deva were going to be partners after all.

As soon as the painful meal was over, Jackson cleared his throat. "I presume you have carefully explained the preliminaries to Miss Warel, Dr. Johns?"

"I've explained nothing!" the woman officer snapped. "Deva Warel states that she absolutely refuses to go through with this arrangement. She had already committed herself to a companion in the University."

“That’s extremely unfortunate,” Bernard Jackson said. “However, in all fairness to my ward, I believe we should go ahead in conformity with established custom.”

“You young people,” he said, “will be assigned quarters in the orientation area. Every opportunity will be offered you to become acquainted and acquire mutual interests. Should this—ah, determination of Miss Warel’s not be overcome we shall consult the Machine for other arrangements. But let us have a fair trial and see what comes of it.”

V

For three days Rick remained in his room without leaving it, except for the brief periods he spent in the restaurant. He wondered how long they would allow the farce to go on, and what would happen to him after it was over.

The fact that Deva Warel was as opposed to the arrangement as he was made it a little easier to bear. But all the humor had gone out of the situation. His heart was sick and empty for all that he had left behind—for the farm, the cattle, even the friends who had refused to help him.

On the fourth day he decided he'd at least have to get out in the open if he was going to preserve his sanity. He left the building and walked out through a large, park-like area enclosed by buildings and masonry walls. He passed occasional couples who were apparently becoming acquainted and "acquiring mutual interests." For the most part they seemed to exhibit a good deal of enthusiasm about the process.

Then he stopped short. A dozen feet away, seated on the grass with a thick book open in her lap, was Deva Warel.

She looked up as he approached and grinned. "Welcome to the bull pen," she said. "I wondered when you were going to come down. Guess I must have really scared you, huh?"

"No, I was just taking my time," he told her. He dropped down beside her on the grass and glanced at the title of the book. *Associative Reflex Circuits of the Memory Package!*

"That's too deep for you, farmboy," she said. "Anyway, since we're here to court, suppose you start courting and get it over with. I've heard they have some real cute customs down on the farm."

"I wouldn't know about that," said Rick slowly. "But ordinarily we use courtesy and kindness to the animals. They're the only creatures below an Agro Sixth Rating, of course."

She looked at him with an expression of regret in her eyes. "I'm sorry. I didn't mean it that way. Or maybe I *did*. I just don't know!" She turned

away again, her expression changing to one of anguish. “You couldn’t know anything about what this means to me. A week ago I hoped and halfplanned to marry a man of my own status—a Mech, First Rating. He’s a genius, and we’ve worked together all our lives.”

“I’m told that’s what you have to expect in return for the benefits of the Machine’s great wisdom. But even an Agro can understand what it means to rebel. I had hopes too—of marrying the woman I loved.”

She looked up quickly, her eyes searching his face. “I didn’t know. That makes us partners in at least one respect, doesn’t it?”

“I guess it does. The question is: What will they do to us for refusing to pair with each other? Will it mean prison, or alteration?”

“I don’t really know. It happens so rarely. I don’t suppose you’d marry me even to escape imprisonment.”

“Not if the future of the whole race depended on it,” Rick said fervently. “An Agro Sixth may not have much to be proud of. But he has a great deal of pride, just the same!”

“I thought so,” said Deva. “That makes the feeling *completely mutual*.”

Abruptly, she got up and moved to a more open spot away from the trees and bushes. She gestured to him to follow her. Irritated, he almost decided to walk back to the building, but his curiosity got the better of him.

“Closer,” she said as he approached. “This place is wired for sound in every bush and pebble. They’re watching us from observation posts on the walls and in the buildings—and they are taking continuous pictures. They’re interested in the progress of our courtship, you see.”

Rick straightened in sudden anger. The whole ordeal was becoming more disgusting by the minute.

“No, be careful!” Deva said. She put her hands up and drew his head closer to her. “Pretend you are whispering endearments in my ear,” she said.

“Keep your face down so they can’t get your lips on the film. They won’t be able to read your words then.”

He did as she advised. She turned her face downward, too, and plucked idly at blades of grass as if they were exchanging only words of love and discussing the most blissful of futures.

“We’ve a chance to get away,” Deva said, “if you have the courage to risk it.”

“I’ve already made one try,” Rick told her. “It failed because I didn’t prepare far enough ahead.” He described his desperate, last-minute attempt to get the plane ready.

Deva looked up in surprise, and then quickly lowered her head again. “You’re a bigger boy than I thought,” she said. “Gerald wouldn’t take the chance either. He had that much in common with Barie.”

“Gerald?”

“The man I worked with and loved.”

Rick looked at her. “Where did you want to go?” he asked. “Tell me, Deva.”

“There are ten thousand square miles of open country outside the cities. They aren’t patrolled very well, because the Machine doesn’t consider the area a vital factor. You’d be surprised how many have escaped and are living in perfect freedom between the cities. Gerald and I could have joined them if he had possessed the nerve—and had loved me enough.”

“Why did he back down?” Rick asked, knowing in advance what her answer would be.

“He said there was no future for us out there. We were civilized citizens of a great community, and had an obligation to do our part. If that obligation included bowing to the superior wisdom of the Machine our only alternative was to obey. Can you imagine a weak-kneed evasion like that from a man who *knows* Mech theory and operation. But of course *you* wouldn’t understand how that knowledge should have made him scorn the Machine. Gerald said he had his work to consider.”

“It hit pretty hard?” said Rick. “Yes—I can see that it did.”

“When he heard I’d been scheduled to pair with an Agro Sixth he acted as if he’d just had a very narrow escape—as if I’d suddenly become unclean. I’d loved him since I was a little girl. But I could have killed him then.”

“You’d take him back, though—if you had the chance?”

Deva nodded. “I’d be fool enough to take him back in a minute if he had the courage to go with me.”

“But he hasn’t,” Rick said, thinking of Barie. “And you haven’t answered my first question. Just what do we do?”

“We can go out there,” Deva said fiercely. “We can help each other escape from the city. After that, we can split up and go our separate ways. I haven’t quite figured out what I’ll do with the rest of my life. But I’d rather be a wilderness outcast than remain chained here, forced to obey every life-denying whim of the Machine.”

“Married to an Agro Sixth, of course.”

Deva nodded. “Married to an Agro Sixth. Do you know what you’ll do? Have you given it serious thought?”

He shook his head, thinking suddenly of all the blank years ahead, without Barie, without home or land or friends. But what Deva said was true. Anything would be better than a life of blind obedience to the whims of the Machine.

“You don’t believe then that the Machine is the infallible god it’s supposed to be?” he asked curiously. “I thought all Mechs bowed to it three times upon arising, and before going to bed.”

Deva’s lips tightened in disgust. “It’s a man-made machine, and only a fool would let it dictate his life for him.”

“There are a good many fools, apparently.”

“Millions of them,” Deva agreed.

“Including Gerald?”

“Including both Gerald and Barie,” said Deva.

Rick started to protest, then grinned. “I think we’ll manage this getaway if it’s physically possible. Just what are your plans?”

“Because you’re an Agro it will be possible for me to get permission to take you out in the city—away from the bull pen. We’ll make several trips, and come back each time. But on the final trip, we’ll just keep going.”

“Is it that easy?”

“There’ll be a search, of course. A search is routine, but it’s never very intensive. After that, however, it’s a long way to the nearest Outlander post and decent food and shelter. You won’t think it’s easy when it’s over.”

“I’m ready to start now,” Rick said, pressing her hand.

She shook her head. “No. Our wisest course is to separate now. Tonight there will be a musical show put on for the benefit of all the gay lovers in the bull pen. You’ll take me to that.”

“Fine. We’ll go on our first tour tomorrow?”

“I’ll see about the arrangements. You’d better pretend to steal a kiss now—just to make it look real.”

Rick complied. After a moment Deva broke away, protesting, with fire in her eyes. “It doesn’t have to be *that* real, farmer-boy! You’ll behave yourself—or find yourself left here alone.”

Rick grinned. “From what you have just told me, you’d have great difficulty in escaping without me. So I don’t intend to worry.”

Even in the sanctuary of his own apartment Rick forced himself to suppress his excitement over the prospects of immediate escape. From what Deva had said, he was certain that the watch circuits were observing him constantly—here as well as out in the park.

He smiled to himself at the memory of Deva’s name for it. The bull pen. Well, what could be more appropriate? He looked down at the strolling couples walking arm-in-arm, as if the whole revolting spectacle had been entirely their idea. In disgust he turned away again. How could men and women make love at the dictates of a machine? What had happened to the human race, anyway?

At least he had to admire and respect Deva. She had courage enough to rebel against the system, despite her certain knowledge that she would lose her place in the only kind of society she knew. And perhaps the price would be death.

Inevitably, his thoughts went back to Barie. He imagined her in a similar position, facing the same kind of challenge. What if the Machine actually should call her up. It could happen!

He went on to imagine her walking down through the gardens and along the flower-scented paths with some man she'd never even seen before, a love choice selected for her by a mindless machine.

The thoughts made him almost physically ill and sharply increased his bitterness.

That night at the entertainment Deva had reassuring news. She had been granted permission to take Rick on extensive tours of the city to acquaint him with the unfamiliar civilization he would soon be expected to embrace as his own. This orientation preliminary was acceptable as part of their courtship period, since he was an Agro.

"But we'll have to wait another four or five days," Deva said. "They want us to become better acquainted in the gardens where there are not too many other distractions. So just be patient," she said as he frowned wryly. "It won't embitter you for life to endure my company for a few extra days!"

She's right, he thought with a grin as he tucked her arm in his. She was going to be real easy to get along with.

The show was not too bad, he guessed. He'd never seen anything quite like it, but it wouldn't matter if he never repeated the experience, either. His mind refused to remain passive. It lunged about, from considerations of his vague and shadowy future to Lonestar and Barie, and then back to the strange little Mech girl sitting beside him. He wondered what would become of her when once they were safely away into the Outlander country. Had she some secret plan of her own?

During the succeeding days they spent most of their time together discussing with each other their past lives in the widely divergent worlds from which they had come. They spent long, idle hours in the gardens or at

the music bar. They ate together and Deva read to Rick poems and stories that opened up a new world of beauty to him.

The time passed more quickly than he could have dreamed. Then on the fifth day Deva said, “Tomorrow we can take a trip through the city. I don’t suppose I’ll need a leash—”

He laughed and drew her to him and kissed her soundly as had become their custom at parting—solely to impress the watchers who recorded all their actions.

“I told you to behave!” Deva said, how angrily he could only guess.

“Tomorrow,” Rick promised.

VI

It was almost like being let out of jail. The administrator, Bernard Jackson, gave them a few brief instructions and showed them the way out of the building.

“Just a kind of once-over-lightly today,” Deva said. “We’ll just go along the streets and through some of the shops to give you an idea of what things are like here.”

“I’d be a lot more interested in seeing—”

“Hush! General sightseeing is *expected* of us. Do you want to ruin everything?”

He found himself fascinated by the sights and sounds of the city in spite of his reluctant interest. The metropolis was built like a machine itself, with precision-placed masses of pleasant-hued steel and concrete flung against the sky and linked with a lacy network of connecting ways up to the highest levels. There was no darkness or filth or visible deterioration anywhere.

“There was a time when big, massed cities like this were considered failures,” Deva said. “It was all the fashion to abandon them, tear them down, and spread the buildings out all over the landscape. That was before we understood the true function of a city. Now we know it can be both a thing of supreme beauty and of maximum efficiency. It’s a device for civilized living. But today’s model compares with a city before the Atom War in about the same way that a starship compares with the first wooden cartwheels.”

“Are you trying to sell me something?” Rick asked.

Deva shook her head. “I’d like to sell it to myself.”

“Why can’t you?”

“Because of the Machine. It’s inexplicable. The Machine actually made possible the cities we have now. And yet because of the Machine I can’t bear to live here any longer. It just doesn’t make sense!”

“Everybody seems happy enough.” Rick said.

And it was true. The people he passed on the street seemed universally content and peaceful—to judge by their expressions. Within a block or two a hundred strangers had smiled and nodded to him. The city was certainly not a place of violent discontent.

“It’s me and you!” said Deva bitterly. “Even Barie is happy to conform to the will of the Machine. But you’re not. Gerald is glad to be free of me because he’s convinced that Utopia has come. And maybe it has! No one lacks food, shelter, luxuries, aggressive outlets. Nothing is lacking—except whatever is tormenting you and me.”

Rick smiled at her outburst. “In the Agro regions you have a chance to become very wise,” he said quietly. “There’s something lacking, all right—something most people don’t even know exists. The love of something that belongs to you alone. You take pride in moulding it to a state of perfection. Pride in yourself because you’re not *owned*.”

“Owned?”

“Of course. All men are property now. Don’t you see? They belong to the Machine and can be dispensed and manipulated as the Machine decides through its all-powerful wisdom. A man cannot have pride if he is owned—and by a mere automaton, at that! He can’t even experience love, because he hasn’t the right to offer himself. He’s no longer the proprietor of his own being.”

Deva walked slowly beside him, her face sober with thought. “I don’t know. That may be the answer. It *would* be strange if it took an Agro to show us what was wrong!”

They walked all day—to the outskirts of the city and back. They stood silently at the edge of the great mass, where the towers dwindled to a thin layer of individual homes. Beyond these stretched the river and the highways leading north and south. Their eyes scanned the distances in silence, each sensing the thoughts of the other, and each saying nothing.

They went more deeply into the heart of the city on the following days. Deva showed Rick the factories and plants that worked in total silence and without attendants, each throwing up a mountain of goods that were

automatically inspected, packed and shipped to locations within the city itself or across the continent without a single human voice to guide them.

On the last day she took him to the headquarters of the Machine. These were many tiers of banked memory circuits and blinking lights below the surface of the city itself, in the depths of the solid rock where nothing short of a convulsion of the Earth could disturb the intricate mechanism.

But there wasn't a great deal to see. And the Machine's more intangible aspects were totally foreign to Rick's mind. They stood in the visitors' gallery, behind the impregnable plastic sheets which afforded them a clear view of the mechanisms and the attendants on the other side.

"There is really no such thing as *the* Machine," Deva said. "In reality there are many hundreds of machines. Every city has one. Its storage vaults contain data on each and every resident. It computes upon that data, and its judgment circuits render decisions based upon its final computations.

"Each machine is interconnected with all others in the nation by tremendous microwave cables. A main clearing panel located somewhere in Kansas links the function of all local machines to a central, coordinating unit.

"In addition, there is a worldwide network of interconnections between New York, London, Paris, Rome, and Moscow. It even extends to Africa and the cities of Central Asia. Actually, for instance, a Mech, First Rating, could be checked out for pairing against a South African native if the Machine so decided. It's never happened as far as I know. But it could happen."

"It's *the* Machine, all right," said Rick bitterly. "I never quite realized before how viciously it has succeeded in wrapping itself around the whole world. You can almost feel it strangling, crushing—"

"Hush!" Deva warned, her fingers tightening in his clasp.

They stood apart from the other visitors to the gallery. But Rick could almost feel invisible eyes and ears ominously recording their every change of expression. He understood Deva's caution.

"There must be some kind of human administration involved in the operation of the Machine," he said more quietly.

Deva nodded. “There is, of course. In the beginning they were the elected representatives of the people—the remnants of the old democratic system.” She tried to keep the bitterness out of her voice. “But now that human safeguard doesn’t even exist any more. The machine picks its own personnel. The original administrators spent most of their time watching and checking on each other to see that no tampering with the mandates occurred. The Machine decided it could do much better by picking men who would not be tempted to tamper.”

The Machine seemed so harmless, with its rows on rows of black panels and glowing tubes and murmuring relays! But it was the enemy. Its very existence had made it forever impossible for him to possess Barie as his own. From every man upon the face of the Earth it had taken something. A desire, a dream, a woman, a treasured possession.

They left the gallery quietly, for they had seen all that they cared to see. They went quickly out and up to the street level and took one of the cars that were freely available to the citizens. Rick glanced at his watch. “It’s almost time to get back to the bull pen,” he said.

Deva did not reply. Her lips were compressed, and she was staring straight ahead of her. Rick looked around at the magnificent towers and airy bridges of the city.

“No regrets?” he whispered.

“No,” Deva said. “No regrets.”

She started the motor of the car, punched the button for the lane and route she desired, and leaned back against the seat with her eyes closed. Swiftly the car moved out into the stream of traffic, and picked up speed.

VII

Rick could not shake off a chill sense of foreboding as the mass of the city was left behind. It did not seem possible they could so easily escape a monster whose tentacles encircled the earth. Might it not even now be watching them through unseen eyes in the walls of the great buildings? When they had gone far enough would it not jerk them back, willing to play a cat-and-mouse game with them until they no longer had energies enough left to continue the fight?

He told himself that he was taking too despairing a view. Deva had lived all her life here, and had been trained as a technical designer of circuits such as those employed by the Machine. She understood its workings, and knew that it could be misled and deceived.

He hoped, too, that someday men would do more than just run away from the Machine. A vague but burning desire had been kindled to activity within him. He would have given his life to be an instrument, no matter how small, in its destruction. But he knew that in his lifetime such a rebellion by the many might well remain only a dream.

He turned his attention to the road as they cleared the advancing edges of the residential section and moved out onto the express highway, heading south.

Deva slowed the car. "We'll be checked at the first automatic station," she said. "The scanners will note that this is a car from the city and record our basic identification. We'll have to abandon the car before then. But that won't be safe, either—until after dark."

His spirits rose as darkness descended and the countryside became more open. The sense of confinement, which had been with him ever since he entered the city, dropped away and he began to feel as free as he had felt during all the years of his childhood in the open country of Lonestar.

"We'll have to turn off soon," said Deva. "Keep on the alert for some kind of trail that might lead down to the river. We'll run the car into the water if we can get it safely down the bank."

A few minutes later, in the glow of the car's headlights, Rick called attention to a sharp turn-off directly ahead. Deva took the controls and in another moment they had come to a halt by the bank of the river.

She directed Rick to get out, and then set the car in motion again. He held his breath as it descended the steep bank. He thought for a moment she had stayed with it too long, but her running figure stumbled to the ground an instant before it struck the water with a resounding splash, and vanished from sight in the swirling dark current.

"Are you hurt?" he called, hastening down the bank.

"My wrist—" she answered in momentary agony. Then, as he reached her side: "I'll be all right. We've got to keep moving as fast as we can."

Her speed and endurance astonished him as they moved through the sheltering brush in the darkness. He wondered how her city life could have endowed her with such stamina. But there was little time for questions.

"There's the check station," she said, a few minutes later. "That red light over on the highway about a half mile down. Our best bet would be to get in the water and float past it. They might have detectors out if our absence has been broadcast."

"Can you swim that far in the dark?" he asked.

"We'll find a driftlog—if we can," she said.

They found a suitable log in the next few minutes. They made bundles of their clothes, so that they could be held out of the water and kept dry as long as possible. Then they slipped soundlessly into the cold, muddy waters, pushing the log forward laboriously between them.

They kept silent, paddling slowly through what seemed an eternity of cold and darkness, letting the current do most of the work. Rick tried to estimate the distance they had traveled, but the light of the inspection station was invisible from the level of the water's surface.

At last Deva spoke, her voice thin with cold and anxiety. "This is far enough. I'm sure we must be well past the station."

They climbed out on the bank once more and saw the signal light a satisfying distance behind them. They dried themselves hastily against the

chill of the night and donned their clothes, which had remained fairly dry.

“Don’t you want to rest for a while?” Rick asked.

“No time for that,” Deva said. “We’ve got to put as much distance as possible between ourselves and that station before daylight. We can find a place to rest and hide all day tomorrow.”

As dawn approached, Deva could no longer conceal her fatigue. They had not slowed during the night, but now they began searching for adequate shelter to protect them from observation during the day. They found it at last in a thick cluster of foliage by the river bank. So dense was its canopy of leaves that even the sky was hidden from their sight when they were at rest in its center.

They ate a small quantity of the food concentrate which they had smuggled out, and almost immediately afterwards Deva fell into a deep sleep.

Rick remained awake until the sun was high overhead, keeping an alert lookout for possible pursuit. There was no sign of a hue and cry from any direction. He began to let himself think that they were no longer in any immediate danger. Possibly the Machine and the Watch Police were not even very much concerned about an escape such as theirs. Citizens who wanted to run away were probably considered of very little value to society.

He allowed drowsiness to creep upon him as he contemplated his future plans. His first and major obligation was to see Deva to the nearest settlement of the Outlanders at which she wished to stay. Then he’d drift on. No place in particular. Eventually, he supposed, he’d have to throw in with the Outlanders somewhere. But until that day arrived—well, he’d always wanted to see as much of the land as possible.

He thought, too, of Deva. She had remarkable courage and an amazing spirit of independence, he admitted in admiration. He might even find himself missing that fiery mop of hair and equally burning temper. In any event, he owed her a debt of gratitude he could never repay.

He was aware next of darkness and of hands tugging at him in ungentle persistence. Deva’s voice whispered impatiently in his ear. “Come on, it’s time we were on our way!”

He struggled awake, and snatched up a food concentrate. He munched on it as they made their way cautiously out of the shelter.

“How much farther must we go before we start hitting Outlander settlements?” he asked.

“It’s hard to tell,” she replied. “They move around. But we should be on the edge of their country by morning.”

They moved rapidly through the wilderness again, following the course of the river, but Deva tired more quickly now. She was willing to stop for rest occasionally but not for long. It was approaching dawn when they heard the first faint thread of unfamiliar sound in the air. Deva paused in alarm and glanced up to the sky, listening.

“What is it?” Rick asked.

“I don’t know. Wait—yes, I do. They’re after us! That’s a helicopter, and it’s heading straight this way.”

Frantically, she raced toward the heavy foliage on the shore. The sound increased and seemed to arrow toward them. They settled in a crouch under the densest cover they could find. But still the sound came on.

Suddenly Deva gasped. “I should have remembered—!”

“Remembered what?”

“Quick!” she urged, grasping his arm. “Into the water!”

Not understanding, Rick stumbled riverward through the brush. “What for?” he asked.

“Infra-red detectors. They’re certain to have them. They may be trained on us already. The water may not mask us. But it’s our only chance.”

They were out of the underbrush now and descending the sloping bank. The sound of the beating vanes was so loud in their ears that it seemed almost upon them. Suddenly a tiny spurt of light snapped out of the sky. Momentarily it cast a ball of flaming brilliance about Deva. Then she cried out and fell to the ground.

“I’m hit! Go on, Rick. Into the water. Forget about me—”

VIII

In his office high in the tower of Orientation Center, Dr. Bernard Jackson put down the report of the Twelfth Sector Watch Police and sighed. He removed his glasses and pinched the bridge of his nose, his features haggard with strain.

“Action completed,” he said finally to Dr. Flora Johns, who occupied a chair nearby. “Everything went precisely according to schedule. I hope it works out. I liked those two. Remember what he called himself and the girl? A Rillo range sire and a salt grass heifer! I’ve got an idea he’ll change his mind about her before long!”

“I fail to see why it was necessary to order the Police to deliberately shoot her,” Flora Johns said with a tartness that verged on bitterness. “She might have been killed. We don’t know yet how serious the injury was. The Police aren’t that good at shooting a moving target from a plane.”

“It was a necessary risk. You know that. You read the Machine’s instructions.”

“But I don’t always believe them!”

“It was necessary because he would have taken her to the first Outlander Settlement, left her there, and gone on his own way. And if he had suggested anything else she would have opposed it.

“It was absolutely necessary to put her in a state of complete dependency on him for a long enough period to enable her to overcome her resistance to his presence. The injury was the only available means.”

“So says that idiot Machine!”

Dr. Jackson smiled.

“Careful, Flora—even the walls have ears. Or so it is said.”

“I don’t care! If the Machine was going to drag us in for insubordination it would have happened long ago. Sometimes I wonder just what the devilish thing is up to, anyway!”

“That’s something I’ve been thinking about for a long time,” he said slowly. “And I think that with this present case I’ve been able to figure it out.”

“What have you figured out?”

“The Machine has recognized from the beginning that we’ve asked it for more than it could deliver. We’ve always assumed that we had succeeded in creating a Machine that could think like a man. We forgot there are two answers to the problem of creating equal thinking in men and machines.

“We haven’t built a Machine that thinks like a man. Instead, we’ve developed a generation of men that think like machines!”

“What are you talking about?”

“Just that. We’ve leveled men and machines to the same category, but not by raising machine thinking. We’ve done it by lowering men. So today we have thinking machines, all right. Millions of them. All over the globe. We call them men!”

“The Machine has its judgment circuits—”

“Fortunately, yes. And this gives it just sufficient ability to recognize its own failure—and wisdom enough to turn the problem back where it belongs.

“What essential difference there is between true human thought, and the kind of thought we’ve developed in the Machine, I don’t know. And I don’t think the Machine knows. But it’s throwing the problem back to the only possible source of solution: human beings. Don’t you see?”

Dr. Flora Johns was staring through the windows of the tower to the glistening city beyond. “If only you were right!” she whispered.

“I have to be right!” Dr. Jackson exclaimed. “It’s the only possible answer. If it weren’t, you and I would have been ordered drawn and quartered long ago. But the Machine picked us—and scores of others like us.

“It keeps us in our present positions and gives out wholly insane instructions—by accepted understanding of its purposes. It gives out pairings like this one we’ve just handled. And provides a clear path for their escape to the Outlander settlements, and insures that they will stay together in spite of themselves. I tell you the Machine is bent on seeing that we

somehow rectify our own mistakes and make men out of those millions of thinking machines that inhabit the world!”

“But to what purpose?” Flora Johns cried. “We’ll be going back to the chaos from which we escaped by the very creation of the Machine. It has showed us how to insure peace, prosperity, happiness. We’ve got the perfection of society that men struggled for thousands of years to obtain!”

Dr. Jackson shook his head.

“No. What we *have* got isn’t it. We may have come close, but for all practical purposes we’ve run up a blind alley. We can thank whatever Providence is watching over us that the Machine has been capable of recognizing that fact.

“Our failure is in turning over final thought and judgment to something outside ourselves. Until we are able to take the responsibility and work out answers with our own personal gray matter, we have not solved the problem. Certainly, our existing solution is a failure.

“Maybe a definite, positive solution is unobtainable. Perhaps the only answer is in the continued searching for an answer. I don’t know, and I feel certain the Machine doesn’t know. But it’s trying to develop a sector of humanity that might be able to find out!”

“It sounds right,” Flora Johns whispered again, and now her eyes were aglow. “It *has* to be right!”

IX

A full harvest season had come and gone, and Rick Theron had stayed with the Outlanders far longer than he had at first had any intention of doing. Deva was on her feet again. She had a little limp, but it had been a long time since her injury had kept her from running and working with the strongest of the Outlander women.

It was time to be moving on.

Yet, as Rick saw her coming from the cabin he'd helped build for her, something caught again deep in the middle of his chest. He wondered what the day would be like that denied him the sight of that red crest of hair, tumbled by the wind, or bound in tender curls.

He moved toward her resolutely.

"You're on your way?" she asked.

"Deva," he said. "I've been doing some thinking. We don't know yet where we're going to end up, and you and I haven't got any better friends out here than each other. We've made out pretty good so far. Why don't we just kind of stick together and see what comes of it?"

She looked at him almost shyly, with a slow smile starting at the corners of her mouth. "I was just about to suggest the same thing," she said.

"You think it could be that the Machine was right about us, after all?"

"It might be wise not to be hasty in going against it," Deva said.

He reached for her suddenly and gathered her into his arms and kissed her long, and passionately.

"Yes," she said. "I think I could even learn to like that. In time and with practice, darling."

Transcriber's Note:

This etext was produced from Fantastic Universe, July 1956 (Vol. 5, No. 6). Extensive research did not uncover any evidence that the U.S. copyright on this publication was renewed.

Obvious errors have been silently corrected in this version, but minor inconsistencies have been retained as printed.

The chapter heading **IX** has been added in this version.

*** END OF THE PROJECT GUTENBERG EBOOK THE THINKING
MACHINE ***

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